

PROJECT DOCUMENT GUIDE

Trustworthy AI for Protein Engineering

Discovering Mutation-Space Reliability and Applying It to Industrial Carbon-Capture Enzyme Design

Master Navigation and Usage Guide for the Complete Project Documentation System

Version 2.0

Purpose Of This Document

The purpose of this document is to explain the entire documentation system of the project.

As the project evolved, the amount of documentation expanded significantly. Different documents now serve different purposes, audiences, and phases of development.

Without a navigation guide, it becomes difficult to understand:

Which document to read first

Which document answers a specific question

How documents relate to one another

Which document is used during development

Which document is used during analysis

Which document is used during publication

Which document is used during competition preparation

This document serves as the master map of the complete project.

Project Overview

This project investigates a fundamental question in trustworthy artificial intelligence.

Can mutation-space sparsity
predict protein stability
prediction failure?

The project seeks to determine whether regions of mutation space that contain less information are more likely to produce unreliable predictions from protein stability prediction systems.

The ultimate goal is not simply to understand prediction failure.

The ultimate goal is to determine whether reliability can be estimated before expensive laboratory testing and then use those reliability estimates to improve protein-engineering decisions.

The complete scientific pathway is:

Mutation Space



Sparsity Framework



Prediction Failure



Reliability Framework



Engineering Decisions



Industrial Carbon-Capture
Applications

Documentation Philosophy

The project documentation follows a layered design.

Each document answers a different category of question.

Examples:

Why are we doing this?

What are we building?

How do the algorithms work?

What data do we need?

How do we implement it?

How do we validate it?

How do we publish it?

How do we compete with it?

No single document attempts to answer every question.

Instead, each document fills a specific role inside the overall ecosystem.

Complete Document Stack

The full project documentation system contains:

1. FINAL PROJECT BLUEPRINT
2. MASTER IMPLEMENTATION SPECIFICATION
3. SCIENTIFIC ALGORITHMS SPECIFICATION
4. DATA ARCHITECTURE &
DATASET SPECIFICATION
5. IMPLEMENTATION ROADMAP
6. CURSOR IMPLEMENTATION PROMPTS

7. SYSTEM OPERATION,
TESTING & VALIDATION HANDBOOK

8. PROJECT MASTER INDEX

9. PROJECT RELEASE PACKAGE

10. PROJECT EXECUTION MANUAL

Together these documents define the entire project.

Document Relationship Map

The documentation hierarchy is:

FINAL PROJECT BLUEPRINT

↓

MASTER IMPLEMENTATION
SPECIFICATION

↓

SCIENTIFIC ALGORITHMS
SPECIFICATION

↓

DATA ARCHITECTURE &
DATASET SPECIFICATION

↓

IMPLEMENTATION ROADMAP

↓

CURSOR IMPLEMENTATION
PROMPTS

↓

SYSTEM OPERATION,
TESTING & VALIDATION

HANDBOOK



PROJECT MASTER INDEX



PROJECT RELEASE PACKAGE



PROJECT EXECUTION MANUAL

Each document depends on the documents above it.

DOCUMENT 1

FINAL PROJECT BLUEPRINT

Purpose

This is the most important scientific document.

The Blueprint explains:

Scientific Motivation

Hypothesis

Background

Project Architecture

Experiments

Reliability Framework

Engineering Application

Primary Question Answered

Why does this project exist?

Intended Audience

Student

Mentor

Research Advisor

Judge

Scientific Reviewer

When To Use

Read first.

Always start with the Blueprint before reading any other technical document.

DOCUMENT 2

MASTER IMPLEMENTATION SPECIFICATION

Purpose

Defines the complete software architecture.

This document explains:

Modules

Services

Dependencies

Pipelines

Execution Flow

System Design

Primary Question Answered

What software must be built?

Intended Audience

Developer

Software Engineer

Research Programmer

When To Use

Read immediately after the Blueprint.

DOCUMENT 3

SCIENTIFIC ALGORITHMS SPECIFICATION

Purpose

Defines every scientific and mathematical calculation.

Topics include:

Empirical Sparsity

Evolutionary Sparsity

Structural Sparsity

Combined Sparsity

Antisymmetry Error

Epistasis Error

Reliability Score

Industrial Fitness Score

Primary Question Answered

How are calculations performed?

Intended Audience

Researcher

Scientist

Algorithm Developer

When To Use

Use whenever implementing or validating calculations.

DOCUMENT 4

DATA ARCHITECTURE & DATASET SPECIFICATION

Purpose

Defines the complete data layer.

Contents:

Schemas

Validation Rules

Keys

Relationships

File Formats

Data Lineage

Primary Question Answered

What data exists and how is it organized?

Intended Audience

Data Engineer

Researcher

Developer

When To Use

Use before creating datasets or writing ingestion code.

DOCUMENT 5

IMPLEMENTATION ROADMAP

Purpose

Defines the order of development.

Contents:

Milestones

Phases

Deliverables

Dependencies

Phase Gates

Primary Question Answered

What should be built first?

Intended Audience

Project Lead

Developer

Research Team

When To Use

Use during planning and development.

DOCUMENT 6

CURSOR IMPLEMENTATION PROMPTS

Purpose

Provides structured implementation prompts for AI-assisted development.

Contains:

Build Prompts

Validation Prompts

Testing Prompts

Refactoring Prompts

Primary Question Answered

How do we instruct Cursor
to build the system?

Intended Audience

Developer

AI-Assisted Programmer

When To Use

Use during implementation.

DOCUMENT 7

SYSTEM OPERATION,

TESTING & VALIDATION HANDBOOK

Purpose

Provides operational instructions.

Contains:

Execution Steps

Testing Procedures

Validation Procedures

Certification Procedures

Failure Recovery

Primary Question Answered

How do we verify the system works?

Intended Audience

Researcher

Developer

Reviewer

When To Use

Use while validating results.

DOCUMENT 8

PROJECT MASTER INDEX

Purpose

Provides project-wide navigation.

Contains:

Document Index

Dataset Index

Figure Index

Scientific Claim Index

Cross References

Primary Question Answered

Where is everything located?

Intended Audience

All Users

When To Use

Use whenever locating information.

DOCUMENT 9

PROJECT RELEASE PACKAGE

Purpose

Defines release and archival procedures.

Contents:

Publication Package

Dataset Package

Source Package

Documentation Package

Knowledge Transfer

Primary Question Answered

How is the project published
and transferred to others?

Intended Audience

Research Lead

Future Researcher

Open-Source Maintainer

DOCUMENT 10

PROJECT EXECUTION MANUAL

Purpose

Defines competition, publication, audit, and long-term strategy.

Contains:

Competition Preparation

Publication Preparation

Judge Questions

Visual Standards

Audit Procedures

Future Research Plans

Commercialization Analysis

Primary Question Answered

How do we maximize scientific,
publication, and competition success?

Intended Audience

Student

Mentor

Publication Reviewer

Competition Judge

Project Lifecycle

The project progresses through five major stages.

Stage 1

Scientific Design

Documents Used:

Blueprint

Algorithms Specification

Stage 2

Software Design

Documents Used:

Implementation Specification

Dataset Specification

Stage 3

Implementation

Documents Used:

Roadmap

Cursor Prompts

Stage 4

Validation

Documents Used:

Validation Handbook

Master Index

Stage 5

Publication And Competition

Documents Used:

Release Package

Execution Manual

Updated Scientific Narrative

The project should always be described using the following progression.

Discovery



Validation



Reliability Estimation



Engineering Application



Industrial Carbon-Capture
Application

Not:

Prediction Model



Mutation Ranking

The scientific contribution is:

Reliability Estimation

The Transketolase and carbon-capture sections are applications of that discovery.

Updated Project Completion Definition

The project is complete only when the following pathway is fully operational.

Empirical Sparsity



Evolutionary Sparsity



Structural Sparsity



Combined Sparsity



Experiment 1



Experiment 2



Integrated Reliability Analysis



Reliability Framework



TKT Mutation Landscape



Active-Site Analysis



Industrial Fitness Framework



Top Candidate Mutations

and all supporting:

Datasets

Figures

Documents

Validation Procedures

Competition Materials

Publication Materials

have been completed and certified.

Recommended Reading Order

For a new researcher, read documents in this exact order:

1. FINAL PROJECT BLUEPRINT

2. PROJECT DOCUMENT GUIDE
 3. MASTER IMPLEMENTATION SPECIFICATION
 4. SCIENTIFIC ALGORITHMS SPECIFICATION
 5. DATA ARCHITECTURE &
DATASET SPECIFICATION
 6. IMPLEMENTATION ROADMAP
 7. CURSOR IMPLEMENTATION PROMPTS
 8. SYSTEM OPERATION,
TESTING & VALIDATION HANDBOOK
 9. PROJECT MASTER INDEX
 10. PROJECT RELEASE PACKAGE
 11. PROJECT EXECUTION MANUAL
-

Final Statement

The purpose of this documentation system is not simply to build a protein-engineering project.

The purpose is to create a complete scientific framework for understanding:

When Artificial Intelligence
Should Be Trusted

within protein engineering and industrial biotechnology.

The central scientific contribution remains:

Mutation-Space Sparsity

↓

Prediction Failure

↓

Reliability Estimation

and the ultimate engineering contribution is:

Reliability-Guided
Protein Engineering

for industrial enzyme design and carbon-capture applications.

END OF PROJECT DOCUMENT GUIDE

Version 2.0

Approved Documentation Architecture for the Complete Trustworthy AI for Protein Engineering Project.